

Propositional Logic

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COMP2711 Discrete Mathematical Tools

1 Formulas and truth tables

1.1 Propositions

MUST UNDERSTAND

Proposition. A declarative sentence with a definite truth value: true or false.

Sentence	Proposition?	Reason
HKUST is in Hong Kong.	yes	It is true.
17 is divisible by 3.	yes	It is false.
$x + 2 = 7$.	no	The value of x is unspecified.
Submit the assignment.	no	It is a command.
Is the server online?	no	It is a question.

Whether a sentence is true is different from whether it is a proposition. A false sentence is still a proposition when its truth value is definite.

MUST UNDERSTAND

Atomic proposition. One complete claim treated as a single unit.

For the door example, define

$card$ = “the card is valid”, pin = “the PIN is correct”, $open$ = “the door opens”.

Each variable names a complete proposition, not an object or number.

1.2 Formulas

MUST UNDERSTAND

Logical formula. An atomic proposition, or propositions combined by logical connectives.

Arithmetic formula	Logical formula
$(x + y) - z$	$(p \vee q) \wedge \neg r$
x, y, z are numbers	p, q, r are T or F
rules: $+, -, \times, \div$	connectives: $\neg, \wedge, \vee, \oplus$

The pattern is the same: variables are combined by rules. In logic, the rules are NOT, AND, OR, and XOR. A proposition variable by itself is also a logical formula.

Build formulas by repeating these rules:

1. Every atomic proposition is a formula.
2. If A is a formula, then $\neg A$ is a formula.
3. If A and B are formulas, then $(A \wedge B)$ and $(A \vee B)$ are formulas.

MUST UNDERSTAND

Connective. An operation that combines one or more formulas to form a new formula.

We first build formulas with $\neg$, $\wedge$, and $\vee$. We define $\oplus$ from these symbols.

Name	Symbol	Read as	True when
negation	$\neg A$	not A	A is false
conjunction	$A \wedge B$	A and B	both are true
disjunction	$A \vee B$	A or B	at least one is true
exclusive disjunction	$A \oplus B$	A XOR B	exactly one is true

$A \vee B$ allows both inputs to be true. Exclusive disjunction, also called **exclusive OR** or **XOR**, does not. It is the derived connective

$$A \oplus B := (A \vee B) \wedge \neg(A \wedge B).$$

The credential condition for the door is now a formula:

$$card \wedge pin.$$

It is true exactly when both credentials are correct. Symbols for the full opening rule come later.

Formula structure

Consider

$$(p \vee q) \wedge \neg p.$$

Its main connective is $\wedge$. The two immediate subformulas are $p \vee q$ and $\neg p$. Read from the inside out: compute the two pieces, then apply $\wedge$.

Parentheses carry meaning. The formulas

$$(p \vee q) \wedge r \quad \text{and} \quad p \vee (q \wedge r)$$

are different: the first requires r ; the second does not.

1.3 Truth tables

A **truth assignment** assigns a truth value to every atomic proposition. With n atoms there are 2^n assignments, because each atom has two choices.

The truth tables make the inclusive–exclusive distinction explicit:

p	$\neg p$	p	q	$p \wedge q$	$p \vee q$	$p \oplus q$
		T	T	T	T	F
T	F	T	F	F	T	T
F	T	F	T	F	T	T
		F	F	F	F	F

Thus $p \vee q$ means “at least one,” while $p \oplus q$ means “exactly one.”

To evaluate $(p \vee q) \wedge \neg p$, compute its subformulas from the inside out:

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$
T	T	T	F	F
T	F	T	F	F
F	T	T	T	T
F	F	F	T	F

The final column is true in exactly one case: p is false and q is true.

Truth-table construction

1. List the atoms and create 2^n input rows.
2. Add a column for each useful subformula.
3. Evaluate inner connectives before outer connectives.
4. Compute the main connective last.
5. Read the answer from the final column.

A truth table is slow but decisive: it checks every possible case.

1.4 Formula classification

MUST UNDERSTAND

Tautology and contradiction. A tautology is true on every assignment; a contradiction is false on every assignment.

Tautologies and contradictions behave like constant arithmetic expressions: their outputs do not depend on their inputs. A **contingency** changes value as its inputs change.

Behavior	Arithmetic example	Logical examples
always 1	$x - x + 1$	tautologies: $p \vee \neg p, q \vee \neg q$
always 0	$x - x$	contradictions: $p \wedge \neg p, q \wedge \neg q$
changes	$x + 1$	contingencies: $p \wedge q, p \vee q$

To show that a formula is not a tautology, one false row is enough. That row is a counterexample. To prove a tautology by truth table, every row must be checked.

1.5 Logical equivalence

MUST UNDERSTAND

Logical equivalence. Two formulas are logically equivalent, written $A \equiv B$, when they have the same truth value on every assignment.

Arithmetic identity	Logical equivalence
$(x + 1)(x - 1) = x^2 - 1$	$A \vee F \equiv A$
same number for every x	same truth value on every assignment

The symbol $\equiv$ is not a connective. It records a judgment that two formulas agree in every possible case.

MUST UNDERSTAND

Equivalence law. A rule that replaces a formula by an equivalent formula without changing its truth value.

If $A \equiv B$, then A may be replaced by B , or B by A , inside a larger formula. The letters A , B , and C may stand for complete formulas, not only atomic propositions.

1.6 Equivalence laws

Equivalence laws make logical calculation possible. They play the same role as algebraic identities: each step changes the expression but preserves its value. The analogy is structural. The symbols $\vee$ and $\wedge$ are not ordinary addition and multiplication, so logic also has laws with no arithmetic counterpart.

Reordering and regrouping

Arithmetic and logic allow the same two basic moves:

Move	Arithmetic	Logic
reorder	$x + y = y + x; xy = yx$	$A \vee B \equiv B \vee A; A \wedge B \equiv B \wedge A$
regroup	$(x + y) + z = x + (y + z)$ $(xy)z = x(yz)$	$(A \vee B) \vee C \equiv A \vee (B \vee C)$ $(A \wedge B) \wedge C \equiv A \wedge (B \wedge C)$

These are the commutative and associative laws. They change order or parentheses, not the propositions themselves.

Distributive laws

The first logical law has the familiar arithmetic pattern

$$x(y + z) = xy + xz, \quad A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C).$$

Logic also has the dual law

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C).$$

In either law, the repeated formula A appears in both terms on the expanded side. Both laws may be used in either direction.

Truth constants

The constants T and F act like special numerical values:

Role	Arithmetic	Logic
neutral value	$x + 0 = x$	$A \vee F \equiv A$
neutral value	$x \cdot 1 = x$	$A \wedge T \equiv A$
absorbing value	$x \cdot 0 = 0$	$A \wedge F \equiv F$
absorbing value	—	$A \vee T \equiv T$

The first pair are identity laws. The second pair are domination laws.

Repetition and opposites

These laws are specific to true-or-false values:

Law	Formulas	Meaning
idempotent	$A \vee A \equiv A; A \wedge A \equiv A$	repeating a claim adds nothing
double negation	$\neg\neg A \equiv A$	two negations cancel
complement	$A \vee \neg A \equiv T; A \wedge \neg A \equiv F$	one is always true; the other is impossible

De Morgan's laws

To negate a compound formula, negate each part and swap AND with OR:

$$\boxed{\neg(A \wedge B) \equiv \neg A \vee \neg B}, \quad \boxed{\neg(A \vee B) \equiv \neg A \wedge \neg B}.$$

The first law says “not both” means “at least one is not.” The second says “neither” means “both are not.”

Absorption and the specialized conditional equivalences appear in the reference table. They are useful rewrites, but they are not part of the core memorization set.

2 Conditionals

2.1 Implication

MUST UNDERSTAND

Implication. The formula $p \rightarrow q$ means that whenever p is true, q must be true. It is read “if p , then q .”

Using an implication

Implications are usually used as rules. Suppose we accept

$$p \rightarrow q$$

and we also know that p is true. Then q must be true; otherwise the accepted rule would be broken.

For example, let

$$p = \text{“20 is divisible by 4”}, \quad q = \text{“20 is even”}.$$

The rule “if a number is divisible by 4, then it is even” gives $p \rightarrow q$. Since p is true, we conclude that q is true.

Evaluating an implication

Sometimes the truth values of both p and q are already known. We then ask whether that particular situation satisfies the rule $p \rightarrow q$.

p	q	$p \rightarrow q$	Does the situation satisfy the rule?
T	T	T	trigger occurs; requirement is met
T	F	F	trigger occurs; requirement is broken
F	T	T	trigger is absent; the rule makes no claim
F	F	T	trigger is absent; the rule makes no claim

There is exactly one false case: p is true and q is false. When p is false, the implication makes no requirement about q , so propositional logic assigns the implication the value T.

This convention can make a strange sentence formally true:

“If the world ended yesterday, then I am an alien.”

The if-part is false, so the implication is true whether or not the conclusion is true. This is a statement about propositional truth value, not evidence of a causal connection or a believable conclusion.

Removing the arrow

An implication can be rewritten using only negation and OR:

$$p \rightarrow q \equiv \neg p \vee q.$$

Both are false exactly when $p = \text{T}$ and $q = \text{F}$.

The first two door requirements can now be written:

$$\text{open} \rightarrow \text{card}, \quad \text{open} \rightarrow \text{pin}.$$

An open door must have both a valid card and a correct PIN.

Necessary and sufficient conditions

In $p \rightarrow q$, p is **sufficient** for q , and q is **necessary** for p .

For example,

$$4 \mid n \rightarrow 2 \mid n.$$

Divisibility by 4 is sufficient for evenness. Evenness is necessary for divisibility by 4. Evenness is not sufficient: 6 is a counterexample.

Converse and contrapositive

Name	General form	Example for p : “It is a dog,” q : “It is an animal”
original	$p \rightarrow q$	A dog is an animal, true
converse	$q \rightarrow p$	An animal is a dog, false: it could be a cat
inverse	$\neg p \rightarrow \neg q$	Not a dog means not an animal, false: it could be a cat
contrapositive	$\neg q \rightarrow \neg p$	Not an animal means not a dog, true

A cat disproves both the converse and the inverse. The contrapositive is equivalent to the original:

$$p \rightarrow q \equiv \neg q \rightarrow \neg p.$$

2.2 Biconditional

MUST UNDERSTAND

Biconditional. The formula $A \leftrightarrow B$ is true exactly when A and B have the same truth value. It is read “ A if and only if B .”

Like every formula, a biconditional receives a truth value only after an assignment v is chosen:

$$v(A \leftrightarrow B) = \text{T} \quad \text{exactly when} \quad v(A) = v(B).$$

Thus the biconditional asks whether A and B agree in the current case. Its answer may be true or false.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

A biconditional is equivalent to two implications:

$$A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A).$$

One direction proves sufficiency; the other proves necessity.

It can also be written without any arrows:

$$A \leftrightarrow B \equiv (A \wedge B) \vee (\neg A \wedge \neg B).$$

The arrow-free form says exactly what the truth table shows: either both formulas are true, or both formulas are false.

We can finally express the original door rule:

$$open \leftrightarrow (card \wedge pin).$$

The left-to-right direction prevents unauthorized opening. The right-to-left direction requires the door to open when both credentials are correct.

Biconditional and logical equivalence

The two symbols have different jobs:

Notation	What it is	What it says
$A \leftrightarrow B$	a compound formula	In one particular case: do A and B agree? Once values are assigned, the formula is T or F.
$A \equiv B$	a judgment about formulas	For every assignment, A and B agree.

Their exact relationship is

$$A \equiv B \quad \text{exactly when} \quad A \leftrightarrow B \text{ is always T.}$$

Here “always” means every assignment to the atomic propositions in A or B . Thus the biconditional is always true exactly when the two formulas agree on every assignment.

Thus $\leftrightarrow$ tests agreement in one case; $\equiv$ asserts agreement in every case.

2.3 Translation

Use this order to translate a statement without reversing an implication:

1. Define short, positive atomic propositions.
2. Mark the structural words: and, or, not, if, only if, iff.
3. For each implication, identify the trigger and the required consequence.
4. Add parentheses, then read the formula back in English.
5. Test a state that the rule should reject.

Useful patterns are:

English	Formula
p if q	$q \rightarrow p$
p only if q	$p \rightarrow q$
p iff q	$p \leftrightarrow q$
neither p nor q	$\neg p \wedge \neg q$
not both p and q	$\neg(p \wedge q)$

Lab door rule

The verbal rule says three things: opening requires a valid card; opening requires a correct PIN; and two correct credentials require opening.

$$(open \rightarrow card) \wedge (open \rightarrow pin) \wedge ((card \wedge pin) \rightarrow open).$$

Combine the two consequences of *open*:

$$(open \rightarrow card) \wedge (open \rightarrow pin) \equiv open \rightarrow (card \wedge pin).$$

The complete rule becomes

$$\begin{aligned} & (open \rightarrow (card \wedge pin)) \wedge ((card \wedge pin) \rightarrow open) \\ & \equiv \boxed{open \leftrightarrow (card \wedge pin)}. \end{aligned}$$

Its truth table identifies every correct state and every system bug:

<i>card</i>	<i>pin</i>	<i>open</i>	$open \leftrightarrow (card \wedge pin)$	Interpretation
T	T	T	T	correct opening
T	T	F	F	functionality failure
T	F	T	F	opens with wrong PIN
T	F	F	T	correctly closed
F	T	T	F	opens with invalid card
F	T	F	T	correctly closed
F	F	T	F	opens without credentials
F	F	F	T	correctly closed

3 Problem solving

3.1 Worked equivalence problem

Show that

$$(p \rightarrow q) \wedge (p \rightarrow \neg q) \equiv \neg p.$$

The task is to prove that the formula on the left and $\neg p$ have the same truth value for every assignment to p and q .

Natural-language reasoning

To read the formula concretely, temporarily let p mean “the door opens” and let q mean “the alarm sounds.” The left side then contains two rules:

- If the door opens, then the alarm sounds.
- If the door opens, then the alarm does not sound.

Suppose the door opens. The first rule requires the alarm to sound, while the second requires it not to sound. These requirements cannot both be met. Therefore, both rules can be true only when the door stays closed, which means p is false.

When the door stays closed, both implications are true regardless of the alarm. Hence the left side is true exactly when $\neg p$ is true. This concrete reading explains the expected answer; we next give two formal proofs for arbitrary propositions p and q .

Two solution methods

Method	What to do
truth table	Check all four assignments and compare the two final columns.
equivalence laws	Rewrite the left side step by step until it becomes $\neg p$.

Truth-table solution

p	q	$\neg q$	$p \rightarrow q$	$p \rightarrow \neg q$	$(p \rightarrow q) \wedge (p \rightarrow \neg q)$	$\neg p$
T	T	F	T	F	F	F
T	F	T	F	T	F	F
F	T	F	T	T	T	T
F	F	T	T	T	T	T

The last two columns match on every row. Therefore,

$$(p \rightarrow q) \wedge (p \rightarrow \neg q) \equiv \neg p.$$

Equivalence-law solution

Use arrow elimination, distribution, the complement law, and the identity law. Start from the left side and change one part at a time:

$$\begin{aligned}
 (p \rightarrow q) \wedge (p \rightarrow \neg q) &\equiv (\neg p \vee q) \wedge (\neg p \vee \neg q) && \text{remove arrows} \\
 &\equiv \neg p \vee (q \wedge \neg q) && \text{distribute} \\
 &\equiv \neg p \vee \mathbf{F} && \text{contradiction} \\
 &\equiv \boxed{\neg p} && \text{remove F.}
 \end{aligned}$$

The second line uses the distributive law from right to left, with $A = \neg p$, $B = q$, and $C = \neg q$. Every line preserves the truth value on every assignment.

Equivalence-law reference

The following tables are a reference. Apply each law from left to right or from right to left.

Law	Equivalent formulas
identity	$p \wedge \mathbf{T} \equiv p$; $p \vee \mathbf{F} \equiv p$
domination	$p \vee \mathbf{T} \equiv \mathbf{T}$; $p \wedge \mathbf{F} \equiv \mathbf{F}$
idempotent	$p \vee p \equiv p$; $p \wedge p \equiv p$
double negation	$\neg\neg p \equiv p$
complement	$p \vee \neg p \equiv \mathbf{T}$; $p \wedge \neg p \equiv \mathbf{F}$
commutative	$p \vee q \equiv q \vee p$; $p \wedge q \equiv q \wedge p$
associative	$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
distributive	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$
absorption	$p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$

Type	Equivalent formulas
conditional	$p \rightarrow q \equiv \neg p \vee q$ $p \rightarrow q \equiv \neg q \rightarrow \neg p$ $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$ $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$ $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$ $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$
biconditional	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$ $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

3.2 Practice

1. Which are propositions: $2^{10} = 1024$; “Close the browser”; x is prime; “Every integer greater than 1 is a product of primes.”
2. Build the truth table for $(p \wedge q) \vee \neg p$ and classify it.
3. Use a truth table to show $\neg(p \vee q) \equiv \neg p \wedge \neg q$.
4. Find the unique assignment that makes $p \rightarrow q$ false.
5. Assume that $p \rightarrow q$ is true and that p is true. What must be true?
6. Translate “the service starts only if the database is available.”
7. Explain the difference between $A \leftrightarrow B$ and $A \equiv B$.
8. Simplify $(p \rightarrow q) \wedge p$.
9. On an island, every inhabitant is either a knight, who always tells the truth, or a knave, who always lies. Let

$$p = \text{“A is a knight”}, \quad q = \text{“B is a knight”}.$$

Let S_A denote the statement made by A, and let S_B denote the statement made by B. The island rule gives

$$p \leftrightarrow S_A, \quad q \leftrightarrow S_B.$$

Determine every possible type assignment in each exchange:

- (a) A says “B is a knight.” B says “A is a knight.”
- (b) A says “B is a knave.” B says “A is a knave.”
- (c) A says “B is a knave.” B says “We are opposite types.”

Answers

1. The first and fourth are propositions; the second is a command; the third has a free variable.
2. The formula is false only when $p = \text{T}, q = \text{F}$, so it is a contingency.
3. The two final columns are F, F, F, T in the standard row order.
4. $p = \text{T}, q = \text{F}$.
5. q must be true; otherwise $p \rightarrow q$ would be false.
6. If s means “service starts” and d means “database available”, then $s \rightarrow d$.
7. $A \leftrightarrow B$ is a formula evaluated on one row; $A \equiv B$ asserts agreement on every row.
8. $(p \rightarrow q) \wedge p \equiv (\neg p \vee q) \wedge p \equiv p \wedge q$.
9. A knight’s statements are true, and a knave’s statements are false. Therefore, a speaker’s type and the truth value of that speaker’s statement agree.
 - (a) Here $S_A = q$ and $S_B = p$, so

$$(p \leftrightarrow q) \wedge (q \leftrightarrow p) \equiv p \leftrightarrow q.$$

The biconditional means that p and q agree, so the possible assignments are

$$(p \wedge q) \vee (\neg p \wedge \neg q).$$

Thus A and B have the same type: both knights or both knaves. The exchange does not determine which of these assignments is the actual one.

- (b) Here $S_A = \neg q$ and $S_B = \neg p$, so

$$(p \leftrightarrow \neg q) \wedge (q \leftrightarrow \neg p) \equiv (p \wedge \neg q) \vee (\neg p \wedge q) \equiv p \oplus q.$$

Thus A and B have opposite types. The exchange again does not determine which person is the knight.

- (c) Here $S_A = \neg q$ and $S_B = p \oplus q$, so

$$(p \leftrightarrow \neg q) \wedge (q \leftrightarrow (p \oplus q)).$$

If p is true, A’s statement forces q to be false. Then $p \oplus q$ is true, which contradicts $q \leftrightarrow (p \oplus q)$. Therefore p is false. A’s false statement forces q to be true. B’s statement $p \oplus q$ is then true, so this assignment is consistent. Thus

$$\boxed{\neg p \wedge q};$$

A is a knave and B is a knight.

4 Summary

The ideas build in this order:

propositions $\longrightarrow$ formulas $\longrightarrow$ truth tables $\longrightarrow$ equivalence
 $\longrightarrow$ implication and biconditional $\longrightarrow$ algebraic reasoning.

The central habit is simple: identify the smallest statements, build the formula, and ask what happens under every possible assignment.